BIN 381 MILESTONE 4

Group P1

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1. Business problem and approach

1.1 The Problem

Households that cannot reliably get enough food are hard to find when support budgets are limited. We worked as analysts for a policy institute of the Department of Social Development type, which must decide which households to prioritise for follow-up and referral. Our evidence is the Stats SA General Household Survey (GHS) 2024, which records 20,940 households and asks eight questions about running out of food, skipping meals and going hungry (Statistics South Africa, 2026).

The project had three business objectives, set in Milestone 1:

- **BO1.** Identify and rank the five to eight household indicators most associated with food insecurity, with $p < 0.05$ and an effect size.
- **BO2.** Build an interpretable classification model, without leakage, that is evaluated on held-out data and can be used to prioritise households.
- **BO3.** Give decision-makers self-service exploration by province and household characteristic, delivered as a Power BI dashboard (Section 5).

We linked these to six hypotheses about what puts a household at risk: falling income (H1), settlement type (H2), large households and low income per person (H3), no electricity (H4), social grant receipt (H5) and poor sanitation (H6). Section 4 reports which were supported.

1.2 How we followed CRISP-DM

We followed the CRISP-DM phases in order (Chapman et al., 2000) and went back when the evidence required it. Table 1 shows where each phase occurred and the most important decision in each.

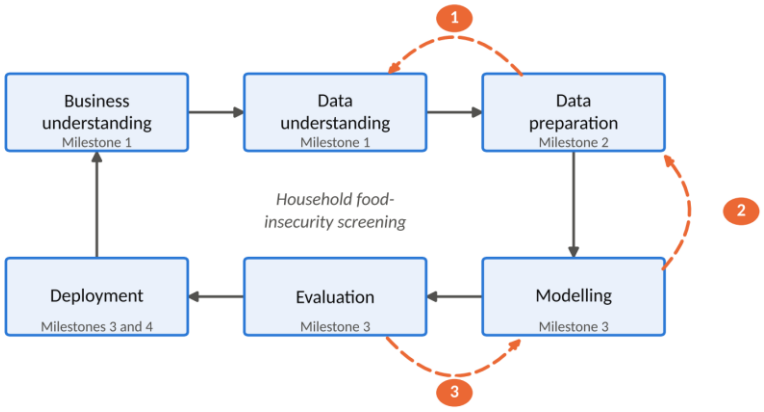


Figure 1 CRISP-DM phases followed in this project. Dashed arrows mark decisions that were revised: (1) duplicate IDs reported in Milestone 1 were a data-type artefact, found in Milestone 2; (2) settlement type recoded to three levels before modelling; (3) three-tier target replaced by the binary fallback for deployment.

Table 1 CRISP-DM phases, where they were done and the main decision in each

Phase	Milestone	Main Decision
Business and data understanding	1	Framed the problem as targeting limited support; explored all 11 GHS files; set BO1 to BO3 and H1 to H6.
Data preparation	2	Kept seven household-level files, repaired keys and duplicates, built the food-insecurity target and the predictors, and fixed one stratified train/test split.

Modelling	3	Tuned three classifiers with stratified cross-validation, then added a binary model when the three-tier target proved too hard.
Evaluation	3	Judged models on imbalance-aware metrics and chose a referral threshold of 0.24 on the training data only.
Deployment	3 and 4	Released an R/Shiny screening tool with validation, monitoring and retraining triggers, and a Power BI dashboard for exploration.

Three times the later work changed an earlier decision. Milestone 1 reported 256 to 272 duplicate household IDs per file, but in Milestone 2 we found this came from reading 18-digit IDs as numbers (Section 2.2). The urban/rural coding of settlement type assumed in Milestone 2 was corrected to the three levels in the data dictionary before modelling. For deployment, we replaced the three-tier target set in Milestone 2 with the binary fallback we documented in advance (Section 3.4).

2. Data strategy, integration and preparation

2.1 Data selected

The GHS 2024 comes as eleven files, D01 to D11. We analysed **households**, so one row is one household, and the key is HouseholdID. Seven household-level files were kept: dwelling conditions (D04), basic services (D05), energy (D06), household well-being (D07, which holds the food-insecurity items and acts as the anchor), connectivity (D08), household resources (D09) and mobility (D10). D08 was not in the Milestone 1 core set. We added it because it is household-level, joins without mismatch, and digital exclusion is a recognised dimension of deprivation.

Three files were left out on purpose. D01 to D03 are person-level (about 70,450 rows against 20,940 households), so using them would force an extra aggregation choice for every variable, and none of H1 to H6 needs a person-level variable. D11 (labour market) comes from a different survey with a different sampling universe and has no household IDs in common with the GHS.

2.2 Cleaning decisions that changed the result

We cleaned each file before joining and logged every treatment (data/clean/qa_log_personC.csv). Table 2 lists the decisions that had a real effect on the final dataset.

Table 2 Main cleaning decisions and their effect

Problem	Decision	Effect
18-digit HouseholdID read as a number loses its last digits (R holds only 15 to 16 digits exactly)	Read every ID as text.	The "256 to 272 duplicates per file" in Milestone 1 disappeared. It was an artefact of rounding, not of the survey.
Exact duplicate rows in D04, D07 and D09	Kept the first copy after checking every duplicate was a full row copy.	20,956, 20,955 and 20,955 rows became 20,940 in each file. No information was lost.
Province typed as padded or mixed-case text (up to 31 variants in one file)	Rebuilt the label from the clean numeric province code.	Nine provinces in every file.
Refused, "unspecified" and "not recorded" codes (9, 99, 9999999 and text tokens)	Recoded to missing instead of treating them as real answers.	For example, 451 households have no usable income. They are imputed later inside the model recipe, using training data only.

Household size of 0	Recoded to missing. Sizes 8 and 9 were kept as valid.	Removes an impossible value without discarding real large households.
Two D09 columns that look like counts but are not (salary earners, total grant recipients)	Excluded from the dataset.	Avoids feeding amounts and sentinels into a count predictor.

2.3 Integration

We joined D07 with the other six files one at a time, using guarded one-to-one joins that stop if a household would be dropped or duplicated. This mattered. Fifteen households in D05 carried placeholder keys ("UNMATCHED_…") instead of real IDs, so a plain inner join would have lost them without any warning. Because all files list households in the same order, we matched each placeholder to the ID at the same row position in D07, and then confirmed it with the household's province, settlement type and survey weight. Both checks agreed for all 15, so the keys were restored and the join went ahead.

The result is 20,940 households with no loss or duplication at any step. We tested it three ways: every source value survives the join unchanged (compared column by column with no tolerance), the descriptors that were dropped from the other files agree with the integrated values for all matched households, and the provincial household counts equal those in D07. These checks stop the notebook if they fail. The dictionary (data/integrated/BIN381_M2_Data_Dictionary.csv) records the source file and treatment of every column.

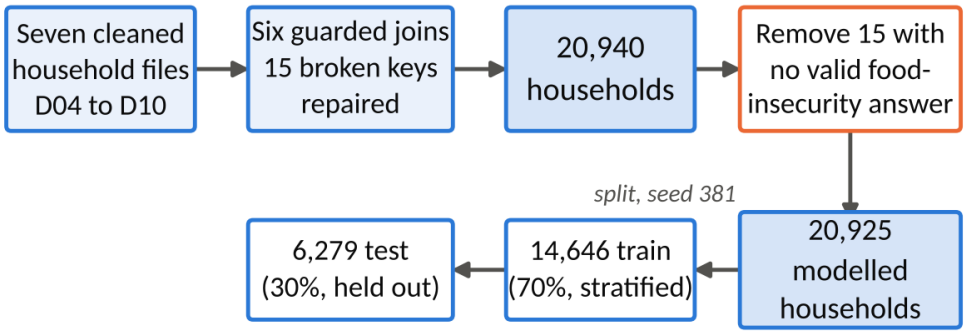


Figure 2 From the cleaned files to the modelling split

2.4 Target, predictors and data split

Target. The Food Insecurity (FI) Score counts the "yes" answers to the eight food-access items (0 to 8). An answer of "don't know/refused" is missing, not "no", because it is no evidence of food security. Fifteen households have no valid answer, so they cannot be scored and were set aside, leaving **20,925 modelled households**. We grouped the score into three tiers: food secure (FI = 0, 69.45% of training households), food insecure (FI 1 to 3, 10.55%) and severely food insecure (FI of 4 or more, 20.00%). Any food insecurity (FI of 1 or more, 30.55%) was kept as a binary fallback.

Predictors. Each of the eight predictors comes from a hypothesis: income compared with a year ago (H1), settlement type (H2), household size and income per person (H3, with log income also used for H1), electricity access (H4), number of grant recipients (H5) and main toilet type (H6). Income is log-transformed because it is strongly right-skewed.

Leakage. The FI Score is built from the eight items, so they cannot also be predictors. We excluded 16 columns: the eight items, two older hunger questions, four "number of days" follow-ups, the FI Score and its count of answered items. The notebook asserts that none is in the predictor set or in the modelling data.

Split. A stratified 70/30 split on the tier (seed 381) gave 14,646 training and 6,279 test households, with the same tier mix in both. The test set was not used for any modelling decision.

3. Modelling

3.1 Framing

This is a supervised classification problem: given a household's income, size, settlement type and access to services and grants, how likely is it to be food insecure? We started with the three-tier target because BO2 asks for vulnerability tiers, and we wrote down in advance that we would fall back to the binary target if the thin middle tier (10.55%) could not be separated.

3.2 Techniques and preprocessing

We built three candidates, one more than the minimum, to cover the trade-off between interpretability and performance. **Regularised multinomial logistic regression** (glmnet; Friedman, Hastie and Tibshirani, 2010) is the interpretable baseline whose coefficients map onto H1 to H6. It assumes a linear relationship on the log-odds scale and low multicollinearity. We handled the first by log-transforming income and checked the second: income compared with a year ago, regressed on the two income features, gives R^2 = 0.064 and a variance inflation factor of 1.07, far below the usual concern level of 5. **A decision tree** (rpart; Therneau and Atkinson, 2026) gives readable rules, but is unstable, so we used it for interpretation and not as the performance candidate. **A random forest** (ranger; Breiman, 2001; Wright and Ziegler, 2017) of 500 trees is the performance benchmark, at the cost of being harder to interpret.

Each family has its own preprocessing recipe. Missing numbers are filled with the training median and missing categories with the training mode, rare toilet codes (under 2% of households) are pooled, and unseen categories are guarded against. The logistic model also gets dummy encoding and scaling, because its penalty is scale-sensitive. All this sits inside tidymodels workflows, so it is re-learned inside every cross-validation fold and never sees the test set (Kuhn and Silge, 2022).

3.3 Validation and tuning

We tuned each model with 5-fold cross-validation on the training set, stratified on the tier so that every fold keeps the 69/11/20 mix. Candidates came from space-filling grids of 8 to 12 settings, and the selection metric was cross-validated ROC-AUC (the Hand and Till, 2001, version for three classes), which the majority class cannot inflate. Accuracy is misleading here because predicting "food secure" for everyone scores 69%.

We also tested class weights as an alternative way to handle imbalance. They raised recall for the severe tier from 25.8% to 58.7% but cut recall for food-secure households from 95.5% to 59.7% and worsened calibration (Brier score 0.289 against 0.205). The app reports a probability, so we kept the unweighted models and handled imbalance at the decision threshold instead (Section 4).

Table 3 Final models and their cross-validated performance (seed 381)

Model	Selected settings	CV ROC-AUC, three tiers	CV ROC-AUC, screening
Multinomial logistic (glmnet)	penalty 8.1×10^{-10}, mixture 0.18	0.669 ± 0.007	0.734
Decision tree (rpart)	cost complexity 1×10^{-10}, depth 11, min_n 19	0.658 ± 0.010	0.721
Random forest (ranger)	mtry 2, min_n 31, 500 trees	0.673 ± 0.005	0.739

Binary logistic (glmnet)	penalty 1.9×10^{-6}, mixture 1 (lasso)	not applicable	0.733 ± 0.010

Screening AUC is the ability to separate food-secure from food-insecure households, from the sum of the two insecure-tier probabilities.

3.4 The change to a binary model

All three three-tier models rank households better than chance (cross-validated AUC 0.66 to 0.67), but on the test set none of them recall more than 3.2% of middle-tier households (the forest 0.3%; Section 4), and the differences between them are small. With these eight predictors, the middle tier cannot be separated from its neighbours, which is the condition we set in advance for the fallback. The Shiny app also returns one probability per household, which suits a two-class model. We therefore fitted a fourth model, a lasso logistic regression on any food insecurity, with the same predictors and recipe. Section 4 compares it with the others on the test set and explains why it was deployed.

3.5 Reproducibility

The seed 381 is set before every random step, and package versions are pinned with renv (R 4.6.1, 225 packages; Ushey and Wickham, 2026). Running the modelling notebook from a clean session reproduces every saved prediction and the parameter table exactly. The fitted workflows, the predictions and an input schema (names, types, valid ranges) were handed to Rudi for evaluation and to Timothy for deployment.

4. Evaluation and key findings

All four candidates were tuned with stratified five-fold cross-validation on the training set and then scored once on the untouched 30% test set (6,279 households, 1,919 of them food insecure). Because the three-tier target could not separate the thin middle tier — no three-tier model recalls more than 3.2% of mildly-to-moderately insecure households, and the best-ranking one manages 0.3% — the deployable question was reframed as a binary screen: is this household at risk of any food insecurity? Models are therefore compared on how well they rank households by risk, not on accuracy, which the 69% food-secure majority would flatter.

Table 4 Screening performance on the test set, each model at its own 80%-recall threshold (source: data/evaluation/evaluation_matrix.csv)

Model	ROC-AUC	Threshold	Recall	Precision	Referred	Brier	ECE
Binary logistic	**0.736**	**0.24**	**80.7%**	**41.9%**	**58.9%**	**0.180**	**0.012**
Random forest	0.745	0.23	79.9%	42.8%	57.1%	0.177	0.012
Multinomial logistic	0.736	0.23	82.4%	41.0%	61.4%	0.180	0.010
Decision tree	0.727	0.19	79.9%	42.2%	57.9%	0.187	0.040

4.1 Why this model and this threshold

The selected model is the lasso (glmnet) binary logistic regression at a referral threshold of 0.24. The threshold was fixed on cross-validation, never on the test set, using a policy rule agreed before evaluation: find at least 80% of at-risk households at the fewest referrals. On the test set it finds 80.7% of at-risk households and 85.7% of the severely food insecure, while referring 58.9% of households at 41.9% precision. The F2-optimal threshold of 0.14 was rejected because it refers 83.8% of all households, and the default 0.5 because it misses 68.9% of at-risk households. The random forest scores marginally higher (AUC 0.745 against 0.736; bootstrap difference +0.009, 95% CI 0.002–0.016), but the logistic model was chosen

for one interpretable odds ratio per factor, equally good calibration and a direct fit with the Shiny app — a gain too small to pay for the loss of explanation in a tool that must justify every referral to an official.

It generalises and it is calibrated: cross-validated AUC 0.733 against 0.736 on test (gap 0.0024, so no meaningful overfitting), Brier 0.180 and expected calibration error 0.012, so the scores are usable as probabilities and not only as a ranking. Survey-weighting the test set moves recall only from 80.7% to 78.3%.

4.2 What drives risk, and what the data says about the hypotheses

Permutation importance ranks the drivers consistently across all three models: income change vs a year ago (0.082), income per person (0.033), number of grant recipients (0.029), household income (log) (0.021). Electricity, settlement type, toilet type and household size add little once income is known. The odds ratios below come from an unpenalised logistic fit on the training data (Person A's tuned penalty was effectively zero), so they describe the deployed model while adding confidence intervals — as associations, not causes.

Table 5 Milestone 1 hypotheses against the final evidence (odds ratios with 95% CI)

Hypothesis	Evidence	Verdict
H1 Falling income raises risk	58.3% vs 18.4–21.2% mid-scale; OR 2.05 (1.62–2.59). But code 1 (much higher) is also 34.8%.	Supported; U-shaped
H2 Rural households more at risk, income held equal	34.8% vs 28.5% urban raw; adjusted OR 0.76 (0.66–0.87).	Not supported; the gap is income and services
H3 Larger households more at risk at fixed income	OR 1.04 (1.01–1.07) per member; raw 26.8% → 42.6%, mostly via income per person.	Supported; weak direct effect
H4 No electricity raises risk	47.0% vs 29.7%; OR 2.07 (1.73–2.48). Only ~5% of households affected.	Supported
H5 Grant receipt lowers risk	Reversed: 37.6% vs 20.6%; OR 1.20 (1.16–1.24) per recipient. Grants are means-tested.	Not supported; targeting, not cause
H6 Toilet type is a useful poverty proxy	Code 5 (4,191 households) 40.9% vs 27.2%; OR 1.31 (1.13–1.52). Importance near zero.	Partly; little predictive value

The gaps matter more than the averages. The screen finds 94.0% of at-risk households earning under R2,500 a month but only 39.2% of those over R11,500, and 61.7% of households with no grant recipients against 88.0% of those with them. By province recall runs from 70.8% (Gauteng) to 89.3% (Eastern Cape). It is weakest exactly where food insecurity is least expected — a reason to keep self-referral open rather than trust a 'lower risk' result.

5. Dashboard and deployment

Two things reached a user. A Power BI dashboard serves the programme manager deciding where to send limited field capacity: it describes the 20,925 surveyed households and lets the manager set the referral threshold against real capacity. An R/Shiny application serves the official assessing a household that was never surveyed, one at a time or in a batch, and shows why. Neither decides eligibility.

5.1 The dashboard answers six decision questions

Because the brief warns against showing every variable, the decision questions were written first and each of the first six pages is titled with the one it answers; three further pages carry the evidence behind them. The two first-iteration dashboards were merged into one report on the final analytical dataset, the Milestone 1 file contributing its layout and its geographic, household and data-lineage pages, rebuilt on tables that live in the repository, and the Milestone 4 file the performance, subgroup and driver pages.

Decision question	Page	What the dashboard shows

1. Where is food insecurity concentrated?	Where the need is	Eastern Cape worst at 43.0%, Limpopo lowest at 13.1%; traditional and tribal areas 35.1%. Drills into settlement type.
2. Which kinds of household are most affected?	Who is affected	Risk falls from 42.9% below R2,500 a month to 11.5% above R11,500, and rises with household size, grant receipt and no electricity.
3. Can we trust the screening tool enough to act on it?	How well the screen works	On 6,279 held-out households it finds 80.7% of at-risk households and 85.7% of the severely insecure, at 41.9% precision, ROC-AUC 0.736.
4. Which households does the screen miss, and where?	Who the screen misses	4 of 22 subgroups fall outside a 10-point tolerance band; weakest above R11,500 (39.2%) and with no grants (61.7%).
5. What should we ask about to spot risk early?	What drives risk	Income change against a year ago carries most of the signal; no electricity roughly doubles the odds.
6. How many households would a given threshold refer?	How well the screen works	A slider moves all three together: 95.9% recall at 83.8% referred (0.14) down to 47.4% at 26.2% (0.40).

Table 6 The six decision questions, the page answering each, and the answer.

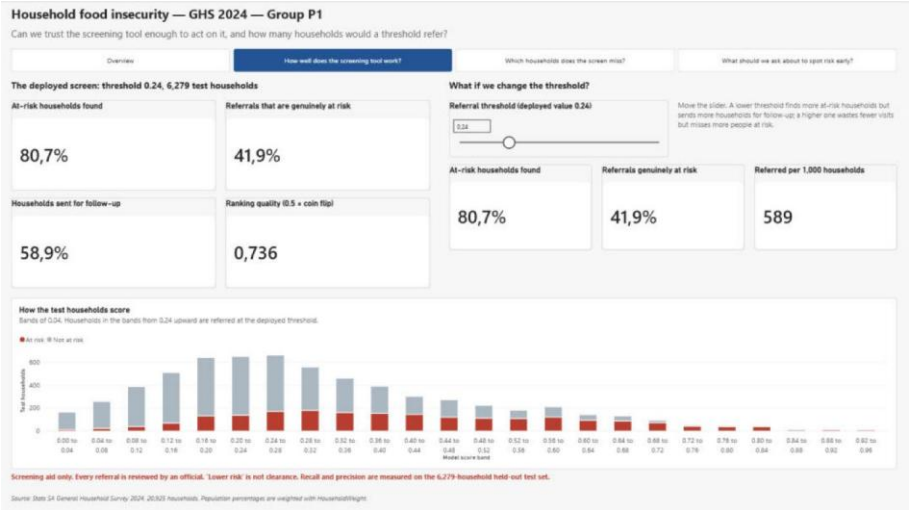


Figure 3 Test-set performance with the threshold what-if.

Four slicers sit in the same position on every household page, every axis is labelled, and no household identifier appears anywhere: the export carries a row number instead. Every page states that this is a screening aid, not an eligibility test. Three further pages carry the evidence: the ten extracts profiled and the seven selected, the 115 data-quality problems found and how each was treated, and the reconciliation in 5.2 below.

5.2 Every dashboard figure reconciles with the analysis

The dashboard recomputes each percentage from the household rows; the R pipeline writes the same percentages independently. A reconciliation page subtracts the two, and all nine provinces, three settlement types and five income bands agree to 0.0 percentage points, nationally at 28.3% at risk, 18.6% severe and 59.2% referred. The what-if panel at the deployed threshold reproduces the headline cards exactly. One distinction is kept visible: 58.9% is the referral rate on the test set, used for every performance claim, while 59.2% is the rate across all 20,925 modelled households, used on the prevalence pages. A build-time check fails if they are swapped.

5.3 The deployed path: validate, clean, preprocess, score

The application serves the same workflow object that was evaluated, at the same threshold of 0.24; both are asserted in code rather than assumed. A typed household and a CSV upload follow one path:

- Validate — the seven inputs must be present, every column is read as text so long household identifiers keep their precision, the eight food-insecurity items are discarded if present (the leakage rule, enforced at the door), and duplicates are flagged.
- Clean — the Milestone 2 rules, column by column: sentinels and text tokens become missing, and unseen codes, impossible values and rows missing more than two inputs are left unscored rather than guessed.
- Preprocess — raw codes become the training features (labelled factors, log income, income per person), then the recipe imputes, encodes and scales using training values only.
- Score — a probability, the words "Higher risk: refer for review" at or above the threshold, and the three inputs that moved it most.

Bad input is a tested case: a 13-row file of deliberately malformed households exercises every rule. Pushing all 6,279 test households back through the app's own path reproduces the saved predictions with a largest difference of zero, so the app, the notebook and the report quote one set of numbers.

5.4 Monitoring and retraining triggers

The outcome is only observed when Stats SA releases a new survey, so between releases the application monitors its inputs and outputs. Each upload appends one row of totals, never household rows, and a monthly review applies the thresholds below.

Signal	Baseline	Alert
Input drift in income, settlement type, household size	PSI 0	0.10 to 0.25 watch; above 0.25 act
Missing income	2.2% at integration	above 5%
Rows not scored	0% of test rows	above 5%
Share referred	58.9%	more than 10 points either way
Recall, at each survey release	0.807	below 0.707
Calibration, at each survey release	Brier 0.180	clearly worse
Subgroup recall, at each survey release	the subgroup table	gap above 10 points

Table 7 Monitoring thresholds, each tied to the test-set baseline.

A simulated shift makes the first row concrete: raising nominal incomes by 25%, roughly five years of inflation, drops the share referred from 59.8% to 51.9% with no change in real need. That is the failure mode this screen is most exposed to, because income enters the model in rands. Retrain when any alert fires, when the survey changes one of the eight predictor items, or annually at each release; retire the tool if recall for any group above 500 households falls below 0.65. Rollback is by model version, recorded with the test metrics that version achieved.

6. Recommendations

Each recommendation below starts from a number in Section 4 and names who should act. All of them assume the screening conditions in Section 5: an official reviews every referral, and the tool never decides eligibility on its own.

1. Deploy the screen for outreach triage only, and resource it for the referral rate it actually produces. At threshold 0.24 the tool refers 58.9% of households and 58.1% of those referrals turn out to be food secure. Programme managers should size follow-up capacity against that rate before go-live, or raise the threshold knowingly and accept lower recall; it is a policy choice, not a technical one. Monitoring should raise an alert if recall on followed-up households falls below 70.7%.

2. Ask every household about income change over the past year at intake. It is the single strongest predictor (AUC drop 0.082, roughly 2.5× the next variable), and 58.3% of households reporting a sharp fall are food insecure. Field staff can collect it in one question, and it should trigger follow-up on its own even before the full screen is run.

3. Treat grant-receiving households as a group to check, not a group to deprioritise. 37.6% of households with grant recipients are food insecure against 20.6% without. Social development should coordinate food-relief outreach with existing grant registers, which is the cheapest route to the households most at risk — while reading the association as evidence of targeting, not of grants causing harm.

4. Direct food-relief and basic-services budgets to the four provinces with the highest weighted risk first. Weighted to the population, 43.0% of Eastern Cape households are at risk, 40.5% in the Northern Cape, 37.1% in North West and 35.5% in Mpumalanga, against 28.3% nationally. Provincial teams can drill into settlement type, income band and grant status within each province on the dashboard.

5. Run a standing audit of 'lower risk' households in the groups where the model is weakest. Recall falls to 39.2% for households earning over R11,500 and 61.7% for households receiving no grants. A monthly sample of lower-risk households in those groups, reviewed by field officers, is the only way the institute will see the cases the screen misses. Keep self-referral open and never present a low score as clearance.

7. Ethics and responsible AI

The brief asks for ethics across the lifecycle rather than as a closing paragraph. Table 8 records where each issue was dealt with; the paragraphs below give the three that shaped the result.

CRISP-DM phase	Issue	What was done
Business understanding	Could be read as an eligibility test	Scope fixed as screening and prioritisation; an official reviews every higher-risk result, and the caveat appears on every page, result and download
Data understanding	Survey data about identifiable households	Seven predictor fields only; identifiers never displayed or returned
Data preparation	The eight food-insecurity items define the target	Excluded from the predictors and asserted in code at every stage, including in the upload validator
Modelling	Imbalance, and accuracy would flatter the model	Stratified cross-validation; recall, precision, PR-AUC and calibration reported instead of accuracy
Evaluation	The best-scoring model was not the most suitable	Interpretable logistic regression chosen over a marginally better random forest, with the trade-off stated
Evaluation	Unequal recall between groups	Equal-opportunity criterion, 10-point tolerance fixed before results were seen; gaps disclosed, no group-specific thresholds
Deployment	Privacy of households scored in production	Uploads processed in memory and not stored; monitoring keeps totals only; access behind sign-in
Deployment	Stigmatising labels	The result reads "Higher risk: refer for review", never "food insecure"
Monitoring	Only referred households generate new labels	A random audit sample of lower-risk households is followed up monthly

Table 8 Ethics decisions by lifecycle phase.

7.1 Privacy, POPIA and human oversight

Data minimisation was applied at every boundary. The model needs seven fields and receives only those; uploads are held in memory for one request and never written to disk; the monitoring log keeps counts and distributions, never a household row; identifiers are not shown, not returned in downloads, and absent from the dashboard. Because a higher-risk result could affect a household's access to support, a human stays in the loop by default, which is also what POPIA section 71 requires of a decision taken solely by automated means.

7.2 Bias, and the gaps we did not close

Equal opportunity was the chosen criterion, because the costly error here is missing a food-insecure household. Of 22 subgroups, 4 fall outside the 10-point tolerance; two are better than average and two are worse. Recall is 39.2% for households reporting more than R11,500 a month and 61.7% for households receiving no grants, against 80.7% overall: these are food-insecure households that do not look poor on the inputs the model has. We did not close the gap with group-specific thresholds, which would mean treating households differently by income on the basis of a proxy target. Instead it is published on the dashboard, repeated in the app, and turned into an operational rule, that a lower-risk result for a household in either group is not a reason to close an assessment. Farm households, 197 in the test set, have an interval too wide to support any claim of parity, so none is made.

7.3 Transparency, and the use of AI tools

Every result carries a probability, the three inputs that moved it most, and a statement of what the tool is not. The explanation method is honest about itself: it varies one input at a time against a typical household, so it shows association, not cause. The same discipline governs the findings, which is why grant receipt is reported everywhere as a marker of targeting rather than a cause, and why the target is described as a proxy built from eight survey answers on a sample rather than a census. Generative AI assistance was used by all three members, is acknowledged in the report, and is recorded use by use in the ethics and AI usage log with what the tool produced, how it was checked and what the team decided. Nothing was accepted on the assistant's word: AI-assisted code was run against the project's own data and compared with figures computed independently. The members remain the authors of the submission and of every decision in Table 8.

8. Limitations and future work

Modest discrimination is the central limitation. An AUC of 0.736 means the screen separates risk only moderately: at the chosen threshold 58.1% of referred households are not food insecure, and 19.3% of at-risk households are missed. The eight predictors describe a household's resources, not its access to food, and the strongest of them is a single self-reported question about income change.

The evidence is one cross-sectional wave of the General Household Survey 2024, so every result is an association rather than a cause; the reversed grant finding (H5) is the clearest case where targeting and effect cannot be separated. The target is itself a proxy built from self-reported items, which carry recall and social-desirability bias, and its three-tier version was abandoned because the middle tier is not separable with these predictors, so the tool cannot grade severity. Training used the unweighted sample (survey-weighting moves recall by 2.4 points); the person-level files D01–D03 and the D11 labour survey were excluded in Milestone 2; and small subgroups such as farms (197 test households) have intervals too wide to judge. Most importantly, the model has never been validated outside this survey: there is no external cohort and no record of what officials found when they followed up.

Future work follows from those gaps, in order of value. First, capture follow-up outcomes so the next model can be trained and validated on what was actually found rather than on a survey proxy. Second, add a second GHS wave to allow within-household change and a genuine test of whether grants protect against food insecurity. Third, bring in per-person income, employment, debt and local food prices, which the current eight predictors stand in for only crudely. Fourth, revisit severity grading once a richer target exists, and re-tune the threshold against the institute's measured follow-up capacity instead of an assumed 80% recall floor.

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Appendix A. AI usage log, Rudi Jan du Plessis (Person B)

Rudi Jan du Plessis (Person B) used Claude, a generative AI assistant made by Anthropic, through the Claude desktop application. The assistant could read and write files in the BIN381 project folder, run code there and operate Power BI Desktop on Rudi's computer, so its output was produced and tested against the project's own data rather than in isolation. Each use is listed below.

Table 9 Use of AI tools in Milestone 4, Person B

ID	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
AI-M4-B1	17 Oct 2026 Rudi (B) Claude (Anthropic)	Project audit and brief. Claude checked the state of the local repository, found that Main Dash.pbix and some raw files existed only on Rudi's computer and in no commit, and set up a separate branch for the Milestone 4 work. It read the official Milestone 4 brief and compared it with the team guideline.	Each file was checked against the full Git history. The brief was read in full against the 100-mark rubric.	The team was warned that Main Dash.pbix had no backup. The A/B/C split and the 10-page limit were confirmed before writing started.
AI-M4-B2	17 Oct 2026 Rudi (B) Claude (Anthropic)	Report sections 4, 6 and 8. Claude wrote build_personB_sections.py, which reads every figure from data/evaluation/ and data/powerbi/ and generates the three sections, so no number is typed into the text by hand.	The script was re-run from a clean state and its output compared with the Milestone 3 report and the evaluation notebook. Length was checked against the 3-page budget.	Rudi reviewed and approved the findings kept, the five recommendations and the wording of the limitations.
AI-M4-B3	17 Oct 2026 Rudi (B) Claude (Anthropic)	Number check. Claude wrote check_numbers.py, which recomputes the headline metrics from the dashboard export, asserts them against data/evaluation/, and fails if easily confused figures are swapped (58.9% vs 59.2% referred; 28.3% vs 28.8% weighted at risk).	Run before and after every change to the report, slides and dashboard. All assertions pass.	Adopted as the team's pre-submission check for any quoted evaluation figure.
AI-M4-B4	17 Oct 2026 Rudi (B) Claude (Anthropic)	Dashboard specification. Claude wrote the hand-over guide docs/powerbi_m4_person_b_pages.md: decision questions, page layouts, DAX for the what-if measures and label columns, and a reconciliation table of expected values. It found that the CSVs imported as text under the en-ZA locale and documented the fix.	Every expected value was recomputed from powerbi_households.csv and matched against evaluation_matrix.csv.	Rudi used the guide to build the dashboard (AI-M4-B5).
AI-M4-B5	17 Oct 2026 Rudi (B) No AI tool	Dashboard data model and first pages, built by Rudi. In Power BI Desktop Rudi loaded the five tables, set the import locale, added the Threshold what-if parameter, the six measures and the Driver, Factor and Direction columns (using the DAX in AI-M4-B4), and built the first version of all three pages: metric cards, threshold slicer, score chart, subgroup slicer, bars and table, driver bars and odds-ratio table.	Checked against the reconciliation table in the specification.	Own work, recorded so that AI-M4-B6 can be read accurately.
AI-M4-B6	17 Oct 2026 Rudi (B) Claude (Anthropic)	Finishing the dashboard. At Rudi's request Claude operated Power BI Desktop. It rebuilt the layouts of Rudi's three pages from code with the same content, added an Overview page with the six decision questions, the what-if cards, the tolerance-band lines, the H1/H4/H5 comparison charts, and titles, navigation and footnotes on every page. Through TMDL view it added supporting measures, columns and number formats. It fixed four errors: the threshold slider had been	Every card and table was compared with data/evaluation/: 80.7% recall, 41.9% precision, 58.9% referred, ROC-AUC 0.736, 20,925 households, 28.3% weighted at risk. The what-if cards were tested at thresholds 0.14, 0.22, 0.24, 0.30 and 0.50 against the specification's	Rudi's original pages are kept as a backup. Rudi is responsible for explaining and demonstrating every visual. Timothy then merged these pages into the group dashboard (AI-M4-C1).

ID	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
		saved at 0.00; the Threshold Value measure returned 0.24000000000000002, so the slider at 0.24 showed 80.4% recall instead of 80.7%; cards showed raw decimals; and table totals summed recall. It took the page screenshots in docs/figures/.	verification table; all matched.	
AI-M4-B7	17 Oct 2026 Rudi (B) Claude (Anthropic)	Slides 5 to 7. Claude drafted the slides, speaker notes, the live demo script and prepared answers to likely defence questions, and later updated the demo script to the dashboard's actual page names.	Every number was checked against data/evaluation/ and the dashboard.	Draft only. Rudi presents slides 5 to 7 and is responsible for defending them.
AI-M4-B8	17 Oct 2026 Rudi (B) Claude (Anthropic)	Review against the brief. Claude checked all Person B deliverables against the rubric and the team guideline and listed the gaps, including that recommendation 4 depends on the province pages of the dashboard. It also corrected an earlier draft of this log that wrongly said the AI had produced no dashboard work.	The gap list was checked against the repository contents and the brief.	Rudi decided which gaps to act on and raised the team-level ones with the group.
AI-M4-B9	17 Oct 2026 Rudi (B) Claude (Anthropic)	Reflection. Claude produced a first draft of Rudi's personal reflection from the project history.	Not applicable: a personal reflection cannot be checked against data.	The submitted reflection is Rudi's responsibility; this entry records that an AI draft existed.

Appendix B. Ethics report and AI usage log

Ethics decisions and use of AI tools across the final synthesis: dashboard, report, presentation and hand-over.

BIN381 SDHSA project, Group P1. Log lead: Timothy Wubbeling (602543), with Alexander Solomon (601638) and Rudi Jan du Plessis (601342). October 2026.

1. Purpose

The brief asks for ethics to be handled throughout the lifecycle rather than once at the end, and for all use of AI to be acknowledged. This log continues the Milestone 3 log in the same format. Section 2 records the ethics decisions taken during Milestone 4, section 3 every use of AI tools, section 4 the acknowledgement statement for the report, and section 5 how the tools were used responsibly. The Milestone 1 to 3 decisions (E3-01 to E3-13, AI-1 to AI-6) remain valid and are not repeated here; this log is read alongside BIN381_M3_Ethics_and_AI_Usage_Log.docx.

2. Ethics decisions (Milestone 4)

ID	Date	Stage	Issue	Decision and mitigation	Owner
E4-01	17 Oct	Presentation	The Milestone 1 dashboard pages showed raw, pre-cleaning figures	The Milestone 1 geographic and household pages were rebuilt on the integrated modelling dataset rather than merged as they were, so no superseded number is presented as final	C
E4-02	17 Oct	Presentation	The Milestone 1 map was bound to a United States shape file and could never have drawn South African provinces	Removed rather than reused; replaced with a sorted province bar chart, and the removal recorded in the dashboard README	C
E4-03	17 Oct	Presentation	Dwelling type and water source were profiled in Milestone 1 but dropped at variable selection	Left off the dashboard, because showing them beside the model's outputs would imply the screen uses them	C
E4-04	17 Oct	Reporting	Two correct referral rates (58.9% on the test set, 59.2% across all households) could look like a	Both kept, each labelled with its population on the dashboard and in the report; a build-time check fails if they are swapped	C

			contradiction, and harmonising them would have been easier		
E4-05	17 Oct	Privacy	A dashboard distributed as a file could carry household-level data	The export carries a row number in place of HouseholdID; no page shows an identifier, and the dashboard ships aggregates only	C
E4-06	17 Oct	Fairness	The subgroup recall gaps were only in the report, where an operational user would not see them	Given a dashboard page of their own, with 95% intervals so small groups cannot be over-read; group-specific thresholds again declined	C
E4-07	17 Oct	Deployment	A dashboard page viewed on its own loses the context that this is a screening aid	The screening-aid statement is in the footer of all seven pages, not only the landing page	C
E4-08	17 Oct	Responsible AI	The dashboard layout and the semantic model were generated by script with AI assistance, which is not obvious from the file	Disclosed in AI-M4-C1 of this log, with every generator script committed to powerbi/M4_Merged_Dashboard/scripts/ so the output can be inspected and rebuilt	C
E4-09	18 Oct	Presentation	The Milestone 1 data-inventory and data-quality figures had become static prose on the merged dashboard, so a reader could not check them	Rebuilt both pages on live tables measured off data/raw/ and the Milestone 2 cleaning log, so every count is traceable; Before and After are carried as text because they mean different things per row and must not be summed	C
E4-10	19 Oct	Reproducibility	The dashboard read its nine tables through a parameter holding an absolute path, so it only worked on the machine that built it and showed errors and empty visuals everywhere else	Each table's data was gzipped and embedded in the model, removing the parameter and all file access, so the file behaves the same for every member; the build now decodes each payload back out and fails rather than ship one that does not round-trip	C
E4-11	19 Oct	Modelling / data	"Don't know" and "refused" answers to the eight food-insecurity questions could have been counted as "no", which would label unsure households as food secure. The model could also have learned from the same questions it was meant to predict.	Those answers were recoded to missing. The 15 households with no valid answer were set aside before the split. Sixteen columns that define or closely proxy the target were kept out of the predictors, with a check that stops the notebook if one appears.	A
E4-12	19 Oct	Evaluation	One headline recall would have hidden that the screen performs far worse for some groups than others, and the obvious patch - a lower threshold for higher-income households - would mean treating households differently by income on the basis of a proxy.	Equal opportunity was chosen as the fairness criterion and recall is measured per subgroup. The gaps are published in section 4 and on the dashboard rather than patched: 39.2% recall above R11,500 and 61.7% with no grant recipients. Recommendation 5 asks for a monthly audit of lower-risk households in exactly those groups instead.	B

Table 10. Ethics decisions in Milestone 4 (A = Alexander Solomon, B = Rudi Jan du Plessis, C = Timothy Wubbeling).

3. Use of AI tools

Timothy Wubbeling (Person C) used Claude, a generative AI assistant made by Anthropic, through the Claude desktop application. The assistant could read and write files in the BIN381 project folder and run code there, so its output was tested against the project's own data rather than produced in isolation. In Milestone 4 it also opened the Power BI project and queried the loaded model directly, which is how the figures in AI-M4-C1 were confirmed.

#	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
AI-M4-C1	17 Oct 2026 Timothy (C) Claude (Anthropic)	Merging the two dashboards. Claude read both .pbix files, mapped what each page contained, and generated a single seven-page Power BI project: the semantic model (8 tables, 32 measures, 6 calculated columns) and the report definition, in my Milestone 1 layout and dark theme. It wrote the generator scripts rather than hand-	The project was opened in Power BI Desktop and refreshed. Every measure was then queried over the local Analysis Services endpoint and compared with data/powerbi/ and data/evaluation/: 20,925 households, 28.3% at risk weighted, 18.6% severe, 59.2% referred, 6,279 test households, recall 0.807,	I accepted the merge. Claude's draft kept a Milestone 1 map bound to a US shape file and I had it removed (E4-02). I have not yet walked all 92 visuals to

		placing visuals, so the grid stays exact and the build is repeatable.	precision 0.419, referred 0.589, ROC-AUC 0.736. The province reconciliation reads 0.0 points for all nine provinces. A pre-build validator checks every field reference against the model.	confirm each renders as intended; that check is mine before I save the .pbix.
AI-M4-C2	17 Oct 2026 Timothy (C) Claude (Anthropic)	The what-if threshold table. Rudi's hand-over documented the recall/precision trade-off as a prose table. Claude computed it for every threshold from 0.05 to 0.90 from the household export and wrote it out as data/powerbi/powerbi_threshold_curve.csv, plus scripts/make_threshold_curve.R to regenerate it after a retrain.	The deployed row reproduces powerbi_model_summary.csv exactly, and all nine rows of Rudi's documented curve match to three decimals (0.14 -> 95.9%, 0.24 -> 80.7%, 0.50 -> 31.1%). The R script carries the same assertions.	Accepted. The shipped CSV was produced by the Python version; the R script has not been run because R is not installed on my machine, and I must execute it once before relying on it.
AI-M4-C3	17 Oct 2026 Timothy (C) Claude (Anthropic)	Report sections 5 and 7. Claude drafted the dashboard-and-deployment and ethics sections as a script (build_personC_sections.py) that reads every figure from data/evaluation/ and data/powerbi/ at build time, in the same pattern as Rudi's section builder, so the prose cannot drift from the notebooks.	Each figure in the output was traced back to its source file. The deployment description was checked line by line against my own Milestone 3 Person C notebook and app/R/validate.R, score.R and monitor.R. The ethics table was checked against the Milestone 3 ethics log.	I reviewed and edited the draft and I am the author of the judgements in it. Claude flagged that the two sections run to about three pages against a 2.5-page budget; the trimming is mine once Alexander's and Rudi's sections are in and the real page count is known.
AI-M4-C4	17 Oct 2026 Timothy (C) Claude (Anthropic)	Slides 8 to 10 and the demo running order, including the messy-upload demonstration and the prepared answers.	Every number on the slides was checked against the report and data/evaluation/. The demo sequence was walked through against the running app.	Draft only. I rewrote the speaker notes in my own words before rehearsal, since I am the one presenting them.
AI-M4-C5	17 Oct 2026 Timothy (C) Claude (Anthropic)	Hand-over and maintenance notes, the dashboard README, and the first draft of this log including the Milestone 4 ethics decisions in section 2.	The monitoring thresholds and retraining triggers were checked against data/evaluation/monitoring_baseline.csv and R/review_monitoring.R. The ethics rows describe decisions I took during the merge.	Reviewed and accepted. The entries describing my own reasoning are mine; Claude wrote them down.
AI-M4-C6	17 Oct 2026 Timothy (C) Claude (Anthropic)	Reflection. Claude produced a first draft of my personal reflection from this session's work and the project history.	Not applicable - a personal reflection cannot be verified against data.	Rewritten by me in my own words. A reflection that I did not write would not be my reflection, and this entry records that a draft existed.
AI-M4-C7	18 Oct 2026 Timothy (C) Claude (Anthropic)	Restoring the Milestone 1 metrics. I asked whether any of my Milestone 1 dashboard metrics had been lost in the merge. Claude audited it, reported that the food-security measures had carried over but that the data-inventory and data-quality metrics had become static text, and rebuilt both as live pages: one table counted off data/raw/ and one reshaped from my own Milestone 2 cleaning log.	Queried against the loaded model and the sources: 10 datasets profiled, 7 selected, 146,626 rows in the selection, 20,940 households after de-duplication, 46 duplicate rows removed, 229 columns profiled, 115 cleaning-log entries across 16 problem types, 113 columns repaired. The generator asserts each of these and fails if they move. Pages 1 to 6 were re-queried and had not changed.	Accepted. Claude also found that its own build script deleted the project's README and scripts folder on every rebuild, and fixed it. The dashboard is now nine pages.
AI-M4-C8	19 Oct 2026 Timothy (C) Claude (Anthropic)	Fixing the dashboard for the rest of the team. We opened the committed dashboard and got errors on every visual and no data. Claude diagnosed it as the absolute path in the DataFolder parameter, which only resolved on the machine that built the file, and replaced the file-based tables with data embedded in the model so there is no path and no parameter.	Tested by copying the dashboard to a folder with no data directory anywhere near it: all nine tables loaded, 20,925 households, and every figure matched the earlier run including the province reconciliation at 0.0 points. The first attempt at embedding failed in Power BI with 'Invalid binary encoding'; Claude traced that to a formatting bug in its own build script, fixed it, and added a check that decodes each embedded table back out and fails the build if it does not match the source file.	Accepted. Claude also found that a long folder path makes Power BI open a blank document instead of the project, with no explanatory error, and wrote that up as a known issue with the advice to clone somewhere short.
AI-M4-A1	14 Oct 2026 Alexander (A)	Milestone 4 team guideline. Claude read the Milestone 4 brief and the repository and	Deliverables and rubric compared line by line with the brief. Statements about	Used as a planning aid. The split was a

	Claude (Anthropic)	produced a Word guideline: deliverables, a proposed A/B/C split, a plan for each task, the marks map, a timeline and likely defence questions.	what was already in the repository were checked against the cloned repository.	proposal, and the group agreed on its own.
AI-M4-A2	14 Oct 2026 Alexander (A) Claude (Anthropic)	Reviewed sections 1 to 3 and gave advice on talking points for slides 1 to 4.	Looked through the previous milestones as well as the rubric and gave talking points.	I looked through the review and made changes to it as well as created the slides by myself. I am responsible for every sentence.
AI-M4-A3	19 Oct 2026 Alexander (A) Claude (Anthropic)	Four figures: a CRISP-DM diagram, a data-flow diagram from the cleaned files to the train/test split, a chart of the three food-insecurity tiers, and a chart of cross-validated ROC-AUC for the four models. Claude wrote the plotting script.	Every value in the figures was compared with Table 10.5 of the Milestone 3 report and the model parameter table. Each image was viewed for errors and overlaps.	Figures 1 and 2 are in the report. The tier chart was left out for space.
AI-M4-A4	19 Oct 2026 Alexander (A) Claude (Anthropic)	Document tidy-up. Claude added the student numbers to the cover, the table and figure captions, converted the reference list to Harvard style, fixed two typos and widened a table column in my Word file.	Cover details checked against our student numbers. Harvard references checked against the original sources' details.	Accepted. I will check the compiled report for consistent referencing with Timothy.
AI-M4-A5	19 Oct 2026 Alexander (A) Claude (Anthropic)	Helped with reviewing my personal reflection and gave improvements.	Not applicable: a reflection cannot be checked against data.	A reflection that I did not write would not be my reflection, and this entry records that Claude did help with improvement.
AI-M4-B1 to B9	17 Oct 2026 Rudi (B) Claude (Anthropic)	Dashboard pages, report sections 4, 6 and 8, slides 5 to 7, and the check that every evaluation figure quoted in a deliverable reconciles with data/evaluation/. Each use is recorded separately in Appendix A of the report.	See Appendix A: every entry names what was checked and against which file.	Listed here so this log covers the whole group; Appendix A carries the detail and is Rudi's to sign off.

Table 11. Use of AI tools in Milestone 4.

4. Acknowledgement statement for the report

Generative AI assistance (Claude, Anthropic) was used during this project. It was used for planning, for drafting and refactoring code, for building the Power BI project and its generator scripts, and for drafting report and presentation text. Every figure it produced was checked against the project's own data files, and every piece of code it wrote was executed and compared with results computed independently. The members of Group P1 reviewed, edited and approved all submitted material, and remain the authors of the analysis, the decisions and the conclusions. Each use is recorded in this log.

5. Responsible use

Three rules governed the use of AI tools, and they are the reason this log can be specific.

- Nothing was accepted on the assistant's word. AI-written code was run against the project's data, and its output compared with a figure computed by another route. The dashboard is the clearest case: the assistant built it, and it was then checked by querying the loaded model and subtracting the R pipeline's own numbers from it.
- The assistant was corrected when it was wrong. It kept a map from the first iteration that was bound to the wrong country's geography (E4-02), and its first draft of report sections 5 and 7 overran the page budget. Both are recorded above rather than quietly fixed.
- Personal and evaluative writing stays personal. The reflection was redrafted by its author (AI-M4-C6), and the ethics decisions in section 2 record judgements the team made, written down with assistance rather than decided by it.

6. Reflection - Timothy Wubbeling (Person C)

Milestone 4 was the first time the whole project had to agree with itself. My job was not new analysis but making the dashboard, the app and the report quote one set of numbers. Two correct referral rates, 58.9% on the test set and 59.2% across all households, could have looked like a mistake; the fix was to label both rather than hide one.

What surprised me was how much of the merge was judgement rather than technique. My Milestone 1 dashboard read the raw files, so bolting its pages on would have put pre-cleaning numbers beside final ones; rebuilding them on the integrated data was slower and the only honest option. I also removed a map I had been proud of, which was bound to a United States shape file and had never drawn South African provinces.

What changed is where I think accuracy comes from. In Milestone 2 I trusted a clean notebook; now I trust a figure only when two independent paths produce it, which is why the dashboard subtracts the R output from itself in front of the user.

What I would improve: I left reconciliation until the dashboard existed. Writing the check first would have caught drift while it was still cheap.

7. Reflection - Alexander Solomon (Person A)

What worked best for this project is that we split it into 3, so each person did a specific section of each milestone. Because I built the modelling and then wrote about it, I could explain every choice. Writing down in Milestone 2 that we would fall back to a binary target if the middle tier couldn't be separated also paid off. When the three-tier models failed to find that tier, we switched without scrambling.

The biggest surprise was how much went wrong in such small, practical places. My leakage checks failed because I assumed the integrated file was already trimmed, when it still had 128 columns. Later, I ran renv::init() in the wrong project folder and had to redo it. Across the milestones, the target and the final model changed more than I expected, and the project grew from an analysis into a tool with monitoring.

I learned how tidymodels recipes keep preprocessing inside each cross-validation fold, and how to work with Git and GitHub as a team. Next time, I would agree on the model's output format with whoever builds the app much earlier, because the three-tier vs binary mismatch cost us time, and I would set up renv on the first day.

8. Reflection - Rudi Jan du Plessis (Person B)

TO CONFIRM - 150 to 200 words on what worked, what surprised him, what changed across the milestones, what he learned and what he would improve.